

Task No.: 2**Date: 03-01-2026**

Title: Develop an Android application to collect student details using Layout Managers and Spinner

Aim:

To design and develop an Android application that uses layout managers, event listeners, and Spinner (String Array) to collect and display student details.

Code:

MainActivity.kt

```
package com.example.madtask3
```

```
import android.content.Intent
```

```
import android.os.Bundle
```

```
import android.widget.*
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    lateinit var etName: EditText
```

```
    lateinit var etRoll: EditText
```

```
    lateinit var spDepartment: Spinner
```

```
    lateinit var spGender: Spinner
```

```
    lateinit var btnSubmit: Button
```

```
    lateinit var btnClear: Button
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
```

```
        setContentView(R.layout.activity_main)
```

```
        etName = findViewById(R.id.etName)
```

```
        etRoll = findViewById(R.id.etRoll)
```

```
        spDepartment = findViewById(R.id.spDepartment)
```

```
        spGender = findViewById(R.id.spGender)
```

```
        btnSubmit = findViewById(R.id.btnSubmit)
```

```
btnClear = findViewById(R.id.btnClear)
```

```
val deptAdapter = ArrayAdapter.createFromResource(
    this,
    R.array.department_array,
    android.R.layout.simple_spinner_item
)
```

```
deptAdapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item)
spDepartment.adapter = deptAdapter
spDepartment.setSelection(0)
```

```
val genderAdapter = ArrayAdapter.createFromResource(
    this,
    R.array.gender_array,
    android.R.layout.simple_spinner_item
)
```

```
genderAdapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item)
spGender.adapter = genderAdapter
spGender.setSelection(0)
```

```
btnSubmit.setOnClickListener {
```

```
    val name = etName.text.toString().trim()
    val phone = etRoll.text.toString().trim()
```

```
    if (name.isEmpty()) {
        Toast.makeText(this, "Please enter Name", Toast.LENGTH_SHORT).show()
        etName.requestFocus()
        return@setOnClickListener
    }
```

```
    if (phone.isEmpty()) {
        Toast.makeText(this, "Please enter Phone Number",
Toast.LENGTH_SHORT).show()
        etRoll.requestFocus()
        return@setOnClickListener
    }
```

```

        if (phone.length != 10) {
            Toast.makeText(this, "Phone number must be 10 digits",
Toast.LENGTH_SHORT).show()
            etRoll.requestFocus()
            return@setOnClickListener
        }

        if (spDepartment.selectedItemPosition == 0) {
            Toast.makeText(this, "Please select Department", Toast.LENGTH_SHORT).show()
            return@setOnClickListener
        }

        if (spGender.selectedItemPosition == 0) {
            Toast.makeText(this, "Please select Gender", Toast.LENGTH_SHORT).show()
            return@setOnClickListener
        }

        val intent = Intent(this, ResultActivity::class.java)
        intent.putExtra("name", name)
        intent.putExtra("roll", phone)
        intent.putExtra("dept", spDepartment.selectedItem.toString())
        intent.putExtra("gender", spGender.selectedItem.toString())
        startActivity(intent)
    }

    btnClear.setOnClickListener {
        clearAllFields()
    }
}

override fun onResume() {
    super.onResume()
    clearAllFields()
}

private fun clearAllFields() {
    etName.text.clear()
    etRoll.text.clear()
    spDepartment.setSelection(0)
    spGender.setSelection(0)
}

```

```

    }
}

```

ResultActivity.kt

```
package com.example.madtask3
```

```
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
```

```
class ResultActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_result)

        val tvResult = findViewById<TextView>(R.id.tvResult)
        val btnBack = findViewById<Button>(R.id.btnBack)

        val name = intent.getStringExtra("name")
        val roll = intent.getStringExtra("roll")
        val dept = intent.getStringExtra("dept")
        val gender = intent.getStringExtra("gender")

        tvResult.text = ""
        Name: $name
        Phone No: $roll
        Department: $dept
        Gender: $gender
        """.trimIndent()

        btnBack.setOnClickListener {
            finish()
        }
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="@android:color/white">

    <TextView
        android:id="@+id/studentInfo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/student_info"
        android:textSize="18sp"
        android:textStyle="bold"
        android:textColor="@android:color/black"
        android:gravity="center"
        android:padding="16dp" />

    <LinearLayout
        android:id="@+id/mainLayout"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="16dp">

        <EditText
            android:id="@+id/etName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="@string/name_hint"
            android:background="@android:drawable/edit_text"
            android:padding="10dp" />

        <EditText
            android:id="@+id/etRoll"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```

android:hint="@string/phone_hint"
android:inputType="phone"
android:maxLength="10"
android:background="@android:drawable/edit_text"
android:padding="10dp"
android:layout_marginTop="10dp" />

```

```

<Spinner
    android:id="@+id/spDepartment"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="@android:drawable/edit_text"
    android:padding="10dp"
    android:layout_marginTop="10dp" />

```

```

<Spinner
    android:id="@+id/spGender"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="@android:drawable/edit_text"
    android:padding="10dp"
    android:layout_marginTop="10dp" />

```

```

<Button
    android:id="@+id/btnSubmit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/submit"
    android:layout_marginTop="16dp" />

```

```

<Button
    android:id="@+id/btnClear"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/clear"
    android:layout_marginTop="10dp" />

```

```

</LinearLayout>

```

```

</LinearLayout>

```

activity_result.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp"
    android:background="@android:color/white">

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:textColor="@android:color/black"
        android:layout_marginBottom="24dp" />

    <Button
        android:id="@+id/btnBack"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Back" />

</LinearLayout>
```

Output:

Name: Yoga Shri P.K
Roll No: 23BCS192

Name: Yoga Shri P.K
Roll No: 23BCS192

Enter Student Name

Enter Phone Number

Select Department

Select Gender

SUBMIT

CLEAR

Yoga Shri P K

6379513909

Computer Science

Female

SUBMIT

CLEAR

Name: Yoga Shri P K
Phone No: 6379513909
Department: Computer Science
Gender: Female

BACK

Result:

An android application using Layouts and Spinners is created and executed successfully.